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Chemistry 12

Lab 2B: Effect of Concentration on Equilibrium Systems

1 Purpose

To explore the effect of concentration change in the following reaction between Iron and Thiocyanate ions:

$$\underset{\text{(yellow)}}{Fe^{3+}} + \underset{\text{(colourless)}}{SCN^-} \rightleftharpoons \underset{\text{(red)}}{FeSCN^{2+}}$$

2 Procedure

See ***Essential Experiments for Chemistry - Lab 12A, Part II (p.165)*** for the procedure. Note that the initial preparation beaker was split across two groups of students.

3 Data/Observations

The initial solution of liquid appeared a light orange. After the addition of each of the reagents, the colour of the solutions in test tubes B, C, and E changed.

Reagent	Stress	Colour observation	Shift
Control (A)		yellow-orange	
KSCN (B)		orange-red	
$FeCl_3$ (C)		orange-red	
KCl (D)		yellow-orange	
NaOH (E)		pale yellow	

Table 1: Reagent additions and colour observations

4 Conclusion

Recall the reaction equation:

$$\underset{\text{(yellow)}}{Fe^{3+}} + \underset{\text{(colourless)}}{SCN^-} \rightleftharpoons \underset{\text{(red)}}{FeSCN^{2+}}$$

In test tubes B and C, when we added KSCN and $FeCl_3$ respectively, we observed a shift towards the products. This is supported by the observed colour change to a redder solution. When the concentration of the reactants was increased, the forward reaction rate increased due to more frequent collisions. The reverse reaction rate remained unchanged as the reverse reaction does not require the individual species to carry forward. More *red* products were being formed from *yellow* reactants, and the system shifted to the right to reach a new equilibrium.

For test tube E, consider the reaction:

$$Fe^{3+} + 3\,OH^- \longrightarrow Fe(OH)_{3(s)}$$

where Iron reacts with Hydroxide to form a precipitate.

When we added NaOH, we observed a shift towards the reactants. This was supported by the observed colour change to a yellower solution. When the NaOH was added, the OH^- ions reacted with the Fe^{3+} ions to form a precipitate. This resulted in a decrease in the concentration of Fe^{3+}. The decrease in concentration resulted in a decrease in collisions, which in turn decreased the forward rate. As the reverse rate does not depend on the concentration of the reactants, it remained unchanged.

More reactants were being produced than products, and the system shifted to the left to reach a new equilibrium.

For test tubes A and D, we observed no shift in the equilibrium. This is supported by the colour of the solution remaining unchanged. For test tube A, no chemicals were added, meaning there was no stress in the system. It follows that there was then no shift in the equilibrium. For test tube D, adding 3KCl did not affect the equilibrium as the K^+ and Cl^- ions do not participate in the reaction. This results in no stress in the system, and thus no shift in the equilibrium.

Our data table can now be fully filled, showing the effects of each of the reagents:

Reagent	Stress	Colour observation	Shift
Control (A)	n/a	yellow-orange	n/a
KSCN (B)	$[SCN^-]\uparrow$	orange-red	$\rightarrow$
$FeCl_3$ (C)	$[Fe^{3+}]\uparrow$	orange-red	$\rightarrow$
KCl (D)	n/a	yellow-orange	n/a
NaOH (E)	$[Fe^{3+}]\downarrow$	pale yellow	$\leftarrow$

Table 2: Reagent additions with filled in information